

Multidimensional Feature Learning Enhancement in IoT Intrusion Detection: An Adaptive Cost-Sensitive Autoencoder and Weighted Ensemble Approach

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Abstract—Internet of Things (IoT) defense against malicious activities heavily depends on intrusion detection systems (IDS). But unbalanced datasets and fast changing network conditions, prevalent in IoT applications, are common challenges for traditional IDS approaches. To address this challenge, in this paper we introduce a novel IDS framework for real-time IoT applications using a cost-sensitive autoencoder (CSAE) based ensemble method to enhance feature learning and IDS performance. In addition to the original data, we employ multidimensional data representation from various autoencoder layers to train machine learning (ML) models (e.g., Decision Trees (DT) and XGBoost (XGB)) which are further augmented by ensemble techniques. Moreover, we propose a strategic pretraining of the autoencoder using a different dataset, which promotes efficient information transfer and system flexibility. Empirical evaluations using two factual IoT datasets demonstrate a significant performance improvement over state-of-the-art IoT intrusion detection systems. Our IDS attains an impressive 99.99% accuracy, in multi-class configurations and a flawless 100% in binary classifications for the CICIOT2023 dataset. Similarly, for the DS2OS dataset, after finetuning the pretrained network, accuracy measures 99.88% in multi-class scenarios and 99.96% in binary classifications. The proposed cost sensitive autoencoder based weighted average (CSAE-WA) ensemble model outperformed six state-of-the-art intrusion detection techniques in a rigorous comparative analysis, demonstrating superior efficacy in addressing dataset imbalance and enhancing robustness in dynamic real-time IoT scenarios.

Index Terms—Internet of Things, Intrusion Detection, Artificial Intelligence, Autoencoder, XGBoost, Decision Tree, Cost-Sensitive Learning, Transfer Learning, Weighted Average Ensemble

I. INTRODUCTION

The widespread deployment of IoT devices has ushered in an era of unprecedented connectivity, yet this presents significant challenges in detecting unauthorized access. The severity of the threat cannot be understated, as vulnerabilities in these devices can expose entire networks to cyber threats. Real-time IDS is essential for safeguarding the dynamic IoT networks, but it often faces the challenges to adapt and effectively handle rare types of attacks. IDS can be classified

into anomaly-based and signature-based types [1]. Anomaly-based IDS identify deviations from normal traffic but often suffer from high false positive rates, while signature-based IDS, despite their effectiveness against known threats, are less capable against uncommon attacks [2], [3].

ML approaches form the backbone of modern IDS for their adaptability [4], yet a single classifier faces limitations in intrusion detection and causes false alarms [5]–[8]. Additionally, there is a pressing need to develop an intelligent IDS that can adeptly learn from imbalanced training samples, which is common in contemporary factual IoT datasets [9], [10], to ensure that all threats are prioritized equally. Another critical challenge is the need for knowledge transfer between different IDS to leverage insights and improve detection capabilities across various systems.

Our research aims to advance intrusion detection in IoT by addressing the abovementioned challenges. To mitigate these problems, this research employs a cost-sensitive autoencoder (CSAE) based feature learning technique, trained on the CICIOT2023 dataset [9]. By leveraging three different data representations: original dataset, bottleneck layer, and second-to-last encoder layer—we build a weighted average (WA) ensemble using XGB and DT. After initial training, the CSAE will be further fine-tuned on the DS2OS dataset [10] followed by WA ensemble to facilitate knowledge transfer and enhance generalization, to boost the IDS's performance in efficiently detecting intrusions. By using a cost-sensitive autoencoder based weighted average (CSAE-WA) ensemble method, and knowledge transfer of IDS, we aim to demonstrate improved intrusion detection accuracy in real-time IoT scenarios. The contribution of the paper can be summarized as follows:

- We propose a novel real-time IDS using a CSAE with the ability of handling imbalanced datasets, leading to enhanced feature learning.
- Utilization of multidimensional data representations from various layers of the CSAE to train ML models like XGB

and DT, followed by WA ensemble techniques.

- Integration of knowledge transfer through adaptation and pretraining on heterogeneous IoT datasets to enhance the IDS's generalization capability and performance.
- Demonstration of improved F1 scores and accuracy rates, highlighting the efficacy of the proposed IDS.

The following sections of this paper will provide an in-depth analysis of the relevant research, followed by a thorough explanation of our approach and the outcomes of our experiments. In conclusion, we will review the main discoveries and potential areas for further investigation.

II. RELATED WORK

Several studies have addressed the challenge of IDS in IoT, employing various ML and deep learning (DL) techniques.

Ensemble Learning Techniques: Zhou et al. [11] explored ensemble learning in IDS, achieving 99.52% to 99.89% accuracy across NSL-KDD, AWID, and CIC-IDS2017 datasets using classifiers such as C4.5, RF, and Forest PA. However, their approach needs improvement in managing large datasets and detecting rare threats. Gao et al. [12] introduced an ensemble adaptive voting algorithm with a MultiTree model. Utilizing classifiers like DT, RF, KNN, DNN and Adaboost on the NSL-KDD dataset, they achieved an accuracy of 85.2%. This study needed to optimize feature selection and data preprocessing to enhance results. Verma and Chandra [13] devised the RePuTE algorithm for IoT attack detection, achieving high accuracy rates on multiple datasets using a soft-voting mechanism.

Deep Learning Algorithms: Shitharth et al. [14] proposed a CSOM-based RMML framework that demonstrated effective performance on multiple datasets, including UNSW-NB 15, NSL-KDD, BoT-IoT IDS, DS2OS, and NSL-KDD. However, their approach did not incorporate a knowledge transfer mechanism. Similarly, Xu et al. [15] developed the FC-Net model employing DNN and CNN classifiers, achieving an accuracy of 97.87%. However, their approach lacked a knowledge transfer mechanism, limiting its adaptability and scalability in dynamic IoT environments where rapid adaptation to new threats is crucial. Furthermore, Neto et al. [9] proposed the CICIOT2023 dataset and Wang et al. [6] developed the DL-BiLSTM algorithm for lightweight IoT attack detection, achieving an accuracy of 93.13% on the CICIOT2023 dataset. But, these methods suffer from high false positive rate and didn't consider the imbalance distribution of training data.

Huma et al. [16] introduced a hybrid deep random neural network and multilayer perceptron model (HDRaNN) that achieved 98.56% accuracy on the imbalanced DS2OS dataset, but they did not consider the F1 score, a significant oversight for evaluating performance on such datasets. Furthermore, Lin et al. [17] proposed an IDS model using LSTM, enhanced with SMOTE and adjusted cost functions to address data imbalance, achieving 96.2% accuracy on the CSE-CICIDS2018 dataset. Additionally, Jose et al. [18] evaluated DNN, LSTM, and CNN models on the CIC-IDS2017 dataset, reaching up to 98.61%

accuracy. However, their research lacked a comparative analysis with other works and omitted crucial data preprocessing step, limiting its robustness and applicability.

Meta-Heuristic, Stacking and Other Approaches: Reddy et al. [7] employed the CatBoost algorithm on the DS2OS dataset, achieving an accuracy of 99.47%. However, their approach resulted in a high false negative rate. Consequently, Khan et al. [19] used a SMOTE-based RF algorithm for intrusion detection, but it was limited to a subset of the dataset, which may not adequately represent the full scope of the data. Moreover, ElDahshan et al. [20] proposed a meta-heuristic algorithm (BGWO) with multiple binary classifiers and meta-heuristic optimization on UNSW-B15 and CICIDS2017 datasets, achieving accuracies up to 99.63%.

Almomani et al. [21] explored stacking ensemble learning using MLP, SVM, NB, and DT classifiers on the UNSW-B15 dataset. Their approach achieved 97.95% accuracy, but a single dataset may not capture the diversity of threats in different network environments. In addition, Sohail et al. [22] introduced INFUSE, an ensemble approach combining SVM, KNN, DT, and RF classifiers with a meta-learner for IDS on the NSL-KDD dataset. The approach achieved accuracies up to 91.6%, but further research is needed to include recent attack categories and other datasets.

These limitations suggest that there is a need of improvement in intrusion detection accuracy to enhance robustness in IoT. Addressing the shortcomings identified in previous research on IoT IDS, we propose a novel framework that tackles critical issues such as imbalanced datasets, high false alarms and inadequate knowledge transfer.

III. METHODOLOGY

This section presents a detailed description of the suggested architecture and it's related components.

A. Dataset Description

In this study, we employ two recent datasets designed specifically for IoT security applications: CICIOT2023 [9] and DS2OS [10]. The CICIOT Dataset 2023 [9] is a comprehensive collection designed to facilitate the development of security analytics for IoT. This dataset encompasses 33 distinct attacks across 105 IoT devices, categorized into seven types: Distributed Denial of Service (DDoS), Denial of Service (DoS), Recon, Web-based, Brute Force, Spoofing, and Mirai, showcasing the spectrum of cyber threats to IoT devices and contains 46 features to capture the intricate details of these scenarios. It includes varied attack methodologies, from flooding (e.g., UDP, ICMP Flood) to more malicious approaches like SQL Injection and GREIP Flood, reflecting a broad range of vulnerabilities in IoT systems. Fig. 1 depicts the data distribution across various attack classes in the dataset, providing a detailed view of the attack type variability and coverage. This information is essential for researchers aiming

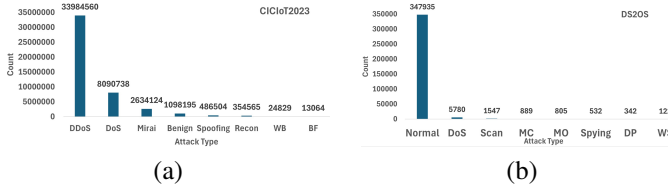


Fig. 1. Overall Dataset Distribution of (a) CICIOT2023, (b) DS2OS

to understand attack prevalence and develop targeted security measures for IoT devices.

Pahl and Aubet introduced the DS2OS dataset [10], a publicly available resource created for the security of next-generation Industrial Internet of Things (IIoT). The dataset contains a total of 357,952 entries, with 10,017 labeled as anomalous and 347,935 as normal as shown in Fig. 1. It encompasses 13 features and includes seven types of attacks: DoS, Malicious Operation, Malicious Control, Wrong Setup, Spying, Scanning, and Data Type Probing attacks.

B. Malicious Entities and Attacker Strategies

Our IDS system can effectively detect the following classes of malicious attacks.

- **Mirai:** Malware that transforms IoT devices into bots for large-scale attacks.
- **DDoS:** Cyber-attack overwhelming target resources from multiple sources, causing unavailability.
- **Brute Force (BF):** Systematically trying every password or encryption key to gain unauthorized access.
- **Spoofing:** Disguising communication to appear as a trusted source for unauthorized access.
- **DoS:** Attack disrupting network or system services, making them unavailable to users.
- **Recon:** Gathering preliminary data to identify vulnerabilities for focused attacks.
- **Web-based (WB):** Exploiting web app vulnerabilities using SQL Injection, XSS, etc.
- **Malicious Control (MC):** Unauthorized access to system devices or networks to extract data.
- **Malicious Operation (MO):** Disrupting system performance and deviating it from intended operations.
- **Spying:** Gaining access to confidential info using system vulnerabilities.
- **Wrong Setup (WS):** Exploiting incorrect system setups for access to sensitive information.
- **Scan:** Scanning network devices to gather IP address and port configuration data.
- **Data Type Probing (DP):** Identifying potential vulnerabilities by methodically scanning systems.

C. Data Preprocessing

The preprocessing stage is crucial for preparing the dataset for subsequent analysis and modeling. Our approach commenced with the consolidation of 169 CSV files from the

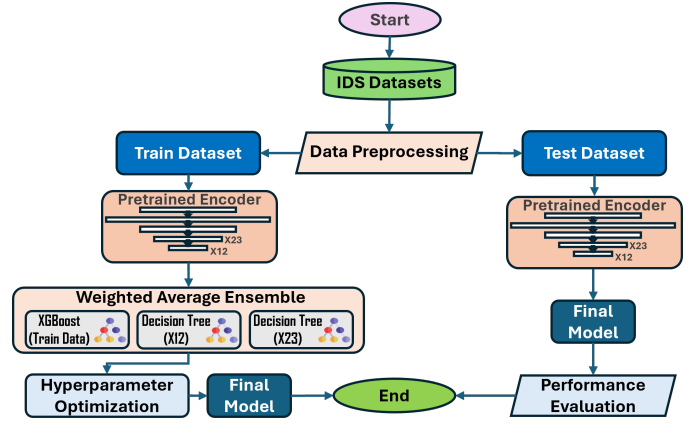


Fig. 2. Proposed Intrusion Detection Architecture

CICIOT2023 dataset repository into a singular DataFrame containing 46 features, ensuring a unified structure for simplified manipulation and analysis. Given the dataset's complexity, with multiple scenarios representing various cyber-attacks, we implemented a categorical simplification to streamline the ML process. This process was facilitated by the `category_extraction` function, which condensed the extensive array of specific attack labels into eight primary categories: DDoS, DoS, Brute Force, Spoofing, Recon, Web-based, Mirai, and Benign Traffic. This categorization effectively transformed the dataset into an eight-class classification challenge, focusing the analysis on the main attack vectors.

For DS2OS Dataset, first we remove redundant and non-informative values in order to make the dataset more efficient. Next, we standardize mixed-type columns into a consistent numeric format and remove columns that have more than 50% missing values and reduce the number of feature to 11 in order to preserve the integrity of the data. Categorical features are transformed using label encoding, and missing values are filled using the k-Nearest Neighbors (kNN) imputation [23] method to optimize the dataset for analysis. To enhance the dataset, we eliminate columns that have constant values and eliminate features that have a correlation coefficient greater than 0.6, thereby reducing multicollinearity.

We applied MinMax scaler to normalize features across both datasets, enhancing the efficiency of subsequent machine learning models. A stratified split was then employed to divide the dataset into training (70%), validation (20%), and testing (10%) sets, ensuring representative class distribution.

D. Cost-sensitive Deep Autoencoder Approach

To address the prevalent issue of class imbalance in the CICIOT2023 dataset, our methodology integrates a cost-sensitive autoencoder. This specialized autoencoder architecture employs class weights inversely proportional to their frequencies, emphasizing the learning of underrepresented attack types.

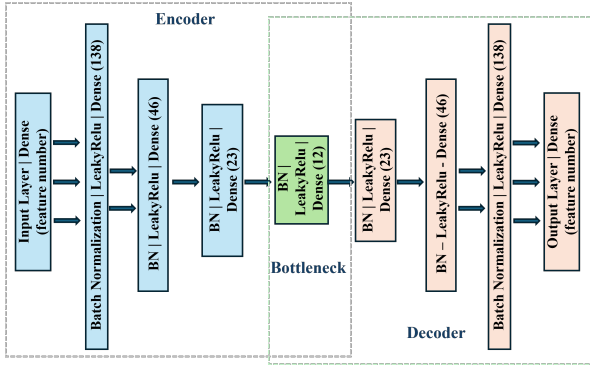


Fig. 3. Architecture of the proposed cost-sensitive autoencoder.

$$class_weights[label] = \frac{total_samples}{count[label]} \quad (1)$$

The architecture consists of an input layer followed by multiple dense layers with increasing and then decreasing units, incorporating LeakyReLU activation and batch normalization (BN) for enhanced stability and performance as shown in Fig. 3. The core of this model is the bottleneck layer, designed for effective feature compression and dimensionality reduction, culminating in an output layer that reconstructs the input where the input size is the same as the feature number in the training dataset.

Algorithm 1 Training Cost-Sensitive Deep Autoencoder on CICIOT2023 Dataset

- 1: Load training and validation sample into DataFrames.
 - 2: Calculate $class_weights[label]$ using Eq. 1.
 - 3: Assign a weight to each sample based on its class.
 - 4: Initialize and compile the model with Adam optimizer and mean squared error (MSE) loss function using Eq. 2.
 - 5: Define ModelCheckpoint for saving the best model and EarlyStopping to prevent overfitting.
 - 6: Fit model on training set and validate on validation set.
 - 7: Save the best model.
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Applying class weights modifies the model's emphasis during training and enhances the effectiveness of learning by balancing the impact of each class. The operational formula for class weights divides the total number of samples by the number of instances of each unique class (e.g., non-attack, type of attack) to guarantee a balanced representation in the loss function, illustrated in Eq. 1 and Eq. 2, respectively.

$$L = \frac{1}{N} \sum_{i=1}^N w_{y_i} (x_i - \hat{x}_i)^2 \quad (2)$$

Where x_i is the input feature vector, $\hat{x}_i$ is the reconstructed feature vector, w_{y_i} is the weight associated with the class of sample i and L represents the calculated loss.

This ensures that underrepresented classes contribute more to the loss function, shown in Fig. 5, guiding the model to learn them better. Implementing the cost-sensitive approach

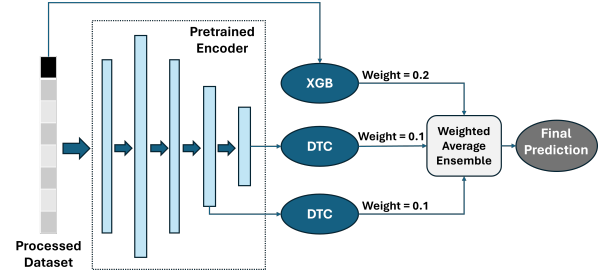


Fig. 4. Proposed CSAE based WA Ensemble Module

outlined in Algo. 1 enables the autoencoder to excel in feature extraction, dimensionality reduction, and managing imbalanced class distribution. This significantly enhances the robustness and accuracy of classification tasks, improving IoT security analytics.

E. Pretraining Autoencoder and Adapting to Heterogeneous Datasets

Our research adapted the CSAE, detailed in Section III-D, from the CICIOT2023 dataset (46 features) to the DS2OS dataset (11 features). We added a new input layer before our pretrained network to upscale DS2OS inputs from 11 to 46 feature representation and added a new output layer after the output of the pretrained network that reproduced the input, enabling transfer of learned features. The autoencoder was fine-tuned to align with DS2OS's characteristics, optimizing its performance. This transfer learning approach, described in Algo. 2, significantly reduced computational effort, demonstrating efficiency in handling datasets with varying feature numbers.

Algorithm 2 Adapting and Pretraining The Cost-Sensitive Autoencoder for the DS2OS Dataset

- 1: Load the existing pre-trained autoencoder.
 - 2: Add input and output dense layers, each with 11 units, to fit the DS2OS dataset into the pre-trained autoencoder.
 - 3: Assign a weight to each sample for DS2OS based on its class, using calculated class weights using Eq. 1.
 - 4: Compile the model with Adam optimizer and MSE loss function using Eq. 2.
 - 5: Define ModelCheckpoint for saving the best model and EarlyStopping to prevent overfitting.
 - 6: Fit model on training set and validate on validation set.
 - 7: Evaluate performance on loss metrics, and save the model.
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F. Weighted Average Ensemble of Machine Learning Models

A WA ensemble [24] approach was used to improve predictive performance by combining three ML models: two DTs [25] and one XGB model [26] as shown in Fig. 4. The DTs, each with parameter criterion of 'gini' and a 'min_samples_split' of 2, utilized features from the last two layers of the encoder, and were assigned a weight of 0.1 each. The XGB model, with a learning rate of 0.3, used

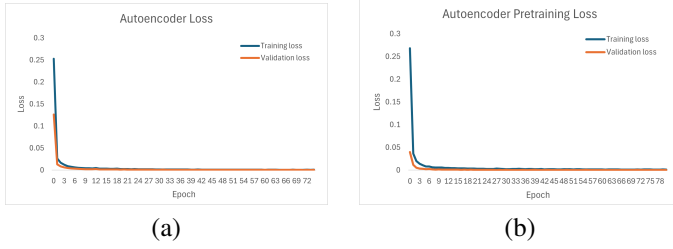


Fig. 5. (a) Training and Validation Loss of the CSAE for CICIOT2023 Dataset, (b) Pretraining Loss for DS2OS Dataset

the original input features and was assigned a weight of 0.2 in the ensemble. After fine-tuning the models, the calibrated ensemble weights allowed for effective combination of their predictions. This balanced the complementary strengths of the DTs and XGB, enhancing classification accuracy and robustness.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

The experiments were implemented in a programming environment using Python 3.10.12 on an Ubuntu 20.04.3 LTS operating system, equipped with 16 GB of RAM and an Intel(R) Xeon(R) CPU running at 2.20 GHz. We utilized additional libraries such as pandas, numpy, sklearn, keras and tensorflow for data processing, ML and DL operations.

B. Evaluation Criteria

The performance of the proposed model is assessed using standard metrics: accuracy, precision, recall, and F1-Score, defined in Eq. 3, 4, 5, and 6. Here, TP (True Positive) and TN (True Negative) indicate correct identifications of attacks and non-attacks, respectively, while FN (False Negative) and FP (False Positive) represent misclassifications of attacks as non-attacks and vice versa.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

To address dataset imbalance, as shown in Fig. 1, our evaluation strategy includes both the F1 score and accuracy. The F1 score, crucial for its balance of precision and recall, is particularly valuable for imbalanced datasets as it accurately measures model effectiveness in identifying positive instances (e.g., attack). While accuracy is straightforward, it can be misleading in scenarios where negative instances (e.g., non-attack) dominate. This dual metric approach ensures a comprehensive evaluation of our model's ability to distinguish between attack and non-attack scenarios.

TABLE I
PERFORMANCE OF PROPOSED METHOD

Dataset	Accuracy	Precision	Recall	F1-score
DS2OS	99.96% (b) 99.88% (m)	100.00% (b) 99.98% (m)	98.60% (b) 99.05% (m)	99.30% (b) 99.50% (m)
CICIOT2023	100% (b) 100% (m)	100% (b) 100% (m)	100% (b) 99.99% (m)	100% (b) 99.99% (m)

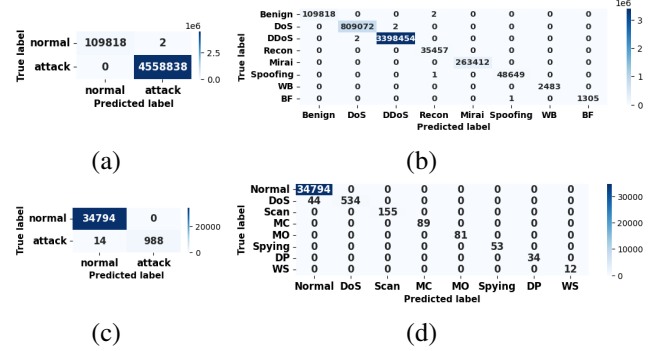


Fig. 6. Confusion Matrix of proposed architecture for (a) CICIOT2023 (binary), (b) CICIOT2023 (8-class), (c) DS2OS (binary), (d) DS2OS (8-class)

C. Results

Our proposed CSAE-WA performed exceptionally well on the CICIOT23 dataset, as illustrated in Table I, with 100% accuracy in binary and multi-class (DoS, Scan, spying, among others) situations, as well as F1-scores of 100% in binary (b) and 99.99% in multi-class (m). In the DS2OS dataset, the model achieved 99.96% accuracy in binary (b) and 99.88% in multi-class (m) scenarios, with F1 scores of 99.30% and 99.50%, respectively. Fig. 6 presents the confusion matrix [27], clearly demonstrating the proposed model's accurate identification of both binary and multi-class attack scenarios. These results underscore the robustness and effectiveness of our approach in accurately classifying both attack and non-attack across diverse classification tasks, thereby validating the choice of these metrics for assessing performance in imbalanced datasets.

Finally, the efficacy of our CSAE-WA model is validated through a comparative analysis with six recent intrusion detection techniques: [6], [7], [9], [14], [16] and [19]. This comparison, detailed in Table II evaluates both accuracy and F1-score. The results clearly indicate that our model outperforms these existing detection techniques in terms of both accuracy and F1- score, thereby confirming its superior efficiency.

V. CONCLUSIONS & FUTURE WORK

This paper has successfully demonstrated the efficacy of a novel real-time IDS tailored for IoT. By integrating a cost-sensitive autoencoder trained on the CICIOT2023 dataset, our CSAE-WA model successfully handled the challenges associated with the imbalanced training samples in dynamic IoT network. The implementation of multidimensional data representations from various layers of the autoencoder followed

TABLE II
PERFORMANCE COMPARISON

Dataset	Article	Model	F1-score	Accuracy
CICIoT2023	Neto et al. (2023) [9]	RF, DNN, MLP, LR, AdaBoost	96.52% (b) 71.92% (m)	99.68% (b) 99.43% (m)
	Wang et al. (2023) [6]	DL-BiLSTM	91.94% (m)	93.13% (m)
	Khan et al. (2024) [19]	RF	99.56% (b) 95.51% (m)	99.56% (b) 95.54% (m)
	Our Method	CSAE-WA	100% (b) 99.99% (m)	100% (b) 100% (m)
DS2OS	Huma et al. (2021) [16]	HDRaNN	98.30% (m)	98.56% (m)
	Reddy et al. (2022) [7]	CatBoost	90.30% (m)	99.47% (m)
	Shitharth et al. (2023) [14]	CSOM-RMML	89.11% (m)	98.76% (m)
	Our Method	CSAE-WA	99.30% (b) 99.50% (m)	99.96% (b) 99.88% (m)

by WA ensemble has significantly enhanced feature learning, which in turn improved the accuracy and robustness of the IDS. Additionally, the strategy of pretraining the autoencoder on a different dataset (DS2OS) facilitated an efficient transfer of knowledge, further enhancing the model's generalization capabilities across diverse IoT scenarios.

On the CICIoT2023 dataset, both binary and multi-class accuracy reached 100%, with F1 scores of 100% and 99.99%, respectively. For the DS2OS dataset, binary accuracy was 99.96% with F1 score of 99.30%, and multi-class accuracy was 99.88% with F1 score of 99.50%. These findings clearly demonstrate that our CSAE-WA model outperforms existing IDSs in both accuracy and F1 score.

In the future, we intend to extend this work to properly tackle unknown attacks. Future research can extend the groundwork established by this study to create more generalized explainable solutions that enhance the capabilities of IDS and ensure the security and integrity of IoT.

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